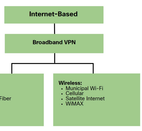
Internet-Based Connectivity

7.6.1

Internet-Based Connectivity Options

Modern WAN connectivity options do not end with Ethernet WAN and MPLS. Today, there are a host of internet-based wired and wireless options from which to choose. Internet-based broadband connectivity is an alternative to using dedicated WAN options.

The figure lists the Internet-based connectivity options.



Internet-based connectivity can be divided into wired and wireless options.

Wired options

Wired options use permanent cabling (e.g., copper or fiber) to provide consistent bandwidth, and reduce error rates and latency. Examples of wired broadband connectivity are Digital Subscriber Line (DSL), cable connections, and optical fiber networks.

Wireless options

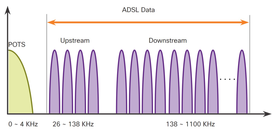
Wireless options are less expensive to implement compared to other WAN connectivity options because they use radio waves instead of wired media to transmit data. However, wireless signals can be negatively affected by factors such as distance from radio towers, interference from other sources, weather, and number of users accessing the shared space. Examples of wireless broadband include cellular 3G/4G/5G or satellite Internet services. Wireless carrier options vary depending on location.

7.6.2

DSL Technology

A Digital Subscriber Line (DSL) is a high-speed, always-on, connection technology that uses existing twisted-pair telephone lines to provide IP services to users. DSL is a popular choice for home users and for enterprise IT departments to support teleworkers.

The figure shows a representation of bandwidth space allocation on a copper wire for Asymmetric DSL (ADSL).



The area labeled POTS (Plain Old Telephone System) identifies the frequency range used by the voice-grade telephone service. The area labeled ADSL represents the frequency space used by the upstream and downstream DSL signals. The area that encompasses both the POTS area and the ADSL area represents the entire frequency range supported by the copper wire pair.

There are several xDSL varieties offering different upload and download transmission rates. However, all forms of DSL are categorized as either Asymmetric DSL (ADSL) or Symmetric DSL (SDSL). ADSL and ADSL2+ provide higher downstream bandwidth to the user than upload bandwidth. SDSL provides the same capacity in both directions.

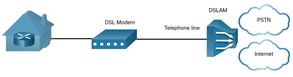
The transfer rates are also dependent on the actual length of the local loop, and the type and condition of the cabling. For example, an ADSL loop must be less than 5.46 km (3.39 miles) for guaranteed signal quality.

Security risks are incurred in this process but can be mediated with security measures such as VPNs.

7.6.3

DSL Connections

Service providers deploy DSL connections in the local loop. As shown in the figure, the connection is set up between the DSL modem and the DSL access multiplexer (DSLAM).



The DSL modem converts the Ethernet signals from the teleworker device to a DSL signal, which is transmitted to a DSL access multiplexer (DSLAM) at the provider location.

A DSLAM is the device located at the Central Office (CO) of the provider and concentrates connections from multiple DSL subscribers. A DSLAM is often built into an aggregation router.

The advantage that DSL has over cable technology is that DSL is not a shared medium. Each user has a separate direct connection to the DSLAM. Adding users does not impede performance, unless the DSLAM Internet connection to the ISP, or to the Internet, becomes saturated.

7.6.4

DSL and PPP

Point-to-Point protocol (PPP) is a Layer 2 protocol that was commonly used by telephone service providers to establish router-to-router and host-to-network connections over dial-up and ISDN access networks.

ISPs still use PPP as the Layer 2 protocol for broadband DSL connections because of the following factors:

- PPP can be used to authenticate the subscriber.
- PPP can assign a public IPv4 address to the subscriber.
- PPP provides link-quality management features.

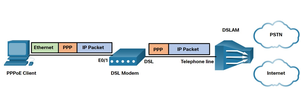
A DSL modem has a DSL interface to connect to the DSL network, and an Ethernet interface to connect to the client device. However, Ethernet links do not natively support PPP.

Click each button for an illustration and explanation of two ways PPP over Ethernet (PPPoE) can be deployed.

Host with PPPoE Client | Router PPPoE Client

Host with PPPoE Client

As shown in the figure, the host runs a PPPoE client to obtain a public IP address from a PPPoE server located at the provider site. The PPPoE client software communicates with the DSL modem using PPPoE and the modem communicates with the ISP using PPP. In this topology, only one client can use the connection. Also, notice that there is no router to protect the inside network.



7.6.5

Cable Technology

Cable technology is a high-speed always-on connection technology that uses a coaxial cable from the cable company to provide IP services to users. Like DSL, cable technology is a popular choice for home users and for enterprise IT departments to support remote workers.

Modern cable systems offer customers advanced telecommunications services, including high-speed Internet access, digital cable television, and residential telephone service.

The Data over Cable Service Interface Specification (DOCSIS) is the International standard for adding high-bandwidth data to an existing cable system.

Cable operators deploy hybrid fiber-coaxial (HFC) networks to enable high-speed transmission of data to cable modems. The cable system uses a coaxial cable to carry radio frequency (RF) signals to the end user.

HFC uses fiber-optic and coaxial cable in different portions of the network. For example, the connection between the cable modem and optical node is coaxial cable, as shown in the figure.

The optical node performs optical to RF signal conversion. Specifically, it converts RF signals to light pulses over fiber-optic cable. The fiber media enables the signals to travel over long distances to the provider headend where a Cable Modem Termination System (CMTS) is located.

The headend contains the databases needed to provide Internet access while the CMTS is responsible for communicating with the cable modems.

All the local subscribers share the same cable bandwidth. As more users join the service, available bandwidth may drop below the expected rate.

7.6.6

Optical Fiber

Many municipalities, cities, and providers install fiber-optic cable to the user location. This is commonly referred to as Fiber to the x (FTTx) and includes the following:

- **Fiber to the Home (FTTH)** – Fiber reaches the boundary of the residence. Passive optical networks and point-to-point Ethernet are architectures that can deliver fiber to homes. Municipal and private FTTH network vary by from on the service provider central office.
- **Fiber to the Building (FTTB)** – Fiber reaches the boundary of the building, such as the basement in a multi-dwelling unit, with the final connection to the individual units being done by copper or coaxial or wireless technology (for example, Wi-Fi).
- **Fiber to the Node/Neighborhood (FTTN)** – Optical cabling reaches an optical node that converts optical signals to a format acceptable for twisted pair or coaxial cable to the premise.

FTTx can deliver the highest bandwidth of all broadband options.

7.6.7

Wireless Internet-Based Broadband

Wireless technology uses the unlicensed radio spectrum to send and receive data. The unlicensed spectrum is accessible to anyone who has a wireless router and wireless technology in the device they are using.

Until recently, one limitation of wireless access has been the need to be within the local transmission range (typically less than 100 feet) of a wireless router or a wireless modem that had a wired connection to the Internet.

Click each button for a description of new developments that are enabling broadband wireless technology.

Municipal Wi-Fi | Cellular | Satellite Internet | WiMAX

Municipal Wi-Fi

Many cities have begun setting up municipal wireless networks. Some of these networks provide high-speed Internet access for free, or for substantially less than the price of other broadband services. Others are for city use only, allowing police and fire departments and other city employees to do certain aspects of their jobs remotely. To connect to a municipal Wi-Fi, a subscriber typically needs a wireless modem, which provides a stronger radio and directional antenna than conventional wireless adapters. Most service providers provide the necessary equipment for free or for a fee, much like they do with DSL or cable modems.

7.6.8

VPN Technology

Security risks are incurred when a teleworker or a remote office worker uses a broadband service to access the corporate WAN over the Internet.

To address security concerns, broadband services provide Virtual Private Networks (VPN) connections to a network device that accepts VPN connections. The network device is typically located at the corporate site.

A VPN is an encrypted connection between private networks over a public network, such as the Internet. Instead of using a dedicated Layer 2 connection, such as a leased line, a VPN uses virtual connections called VPN tunnels. VPN tunnels are routed through the Internet from the private network of the company to the remote site or employee host.

The following are several benefits to using VPN:

- **Cost savings** – VPNs enable organizations to use the global Internet to connect remote offices, and to connect remote users to the main corporate site. This eliminates costly long-distance dedicated connections and modem banks.
- **Security** – VPNs provide the highest level of security by using advanced encryption and authentication protocols that protect data from unauthorized access.
- **Scalability** – Because VPNs use the Internet infrastructure within ISPs and devices, it is easy to add new users. Corporations can add large amounts of capacity without adding significant infrastructure.
- **Compatibility with broadband technology** – VPN technology is supported by broadband service providers such as DSL and cable. VPNs allow mobile workers and telecommuters to take advantage of their home high-speed Internet service to access their corporate networks. Business-grade, high-speed broadband connections can also provide a cost-effective solution for connecting remote offices.

VPNs are commonly implemented as the following:

- **Site-to-site VPN** – VPN settings are configured on routers. Clients are unaware that their data is being encrypted.
- **Remote Access** – The user is aware and initiates remote access connection. For example, using HTTPS in a browser to connect to your bank. Alternatively, the user can run VPN client software on their host to connect to and authenticate with the destination device.

Note: VPNs are discussed in more detail later in this course.

7.6.9

ISP Connectivity Options

Click each button for an illustration and explanation about the different ways an organization can connect to an ISP. The choice depends on the needs and budget of the organization.

Single-homed | Dual-homed | Multihomed | Dual-multihomed

Single-homed

Single-homed ISP connectivity is used by organization when Internet access is not crucial to the operation. As shown in the figure, the client connects to the ISP using one link. The topology provides no redundancy. This is the least expensive solution of the four shown.

7.6.10

Broadband Solution Comparison

Each broadband solution has advantages and disadvantages. The ideal solution is to have a fiber-optic cable directly connected to the client network. Some locations have only one option, such as cable or DSL. Some locations only have broadband wireless options for Internet connectivity.

If there are multiple broadband solutions available, a cost-versus-benefit analysis should be performed to determine the best solution.

Some factors to consider include the following:

- **Cable** – Bandwidth is shared by many users. Therefore, upstream data rates are often slow during high-usage hours in areas with over-subscription.
- **DSL** – Limited bandwidth that is distance sensitive (in relation to the ISP central office). Upload rate is proportionally lower compared to download rate.
- **Fiber-to-the-Home** – This option requires fiber installation directly to the home.
- **Cellular/Mobile** – With this option, coverage is often an issue, even within a small office or home office where bandwidth is relatively limited.
- **Municipal Wi-Fi** – Most municipalities do not have a mesh Wi-Fi network deployed. If it is available and in range, then it is a viable option.
- **Satellite** – This option is expensive and provides limited capacity per subscriber. Typically used when no other option is available.

7.6.11

Lab - Research Broadband Internet Access Technologies

In this lab, you will complete the following objectives:

- Part 1: Investigate Broadband Distribution
- Part 2: Research Broadband Access Options for Specific Scenarios

Research Broadband Internet Access Technologies